Linear Differential Eq^n of Second And Higher Order:-

A differential eq^n is said to be linear in dependent variable if the dependent variable and all its derivatives appear only in the first degree and are not multiplied together.

Thus, a linear differential eq^n of order n is

$$P_0(x)\frac{d^n y}{dx^n} + P_1(x)\frac{d^{n-1}y}{dx^{n-1}} + P_2(x)\frac{d^{n-2}y}{dx^{n-2}} + \dots + P_n(x)y = R(x) \quad \text{---(1)}$$

where the coefficients $P_0, P_1, P_2, \dots, P_n$ and R are assumed to be real functions of x, need not be linear in $[a,b]$ and $P_0(x) \neq 0$ in this interval.

e.g. (1) $e^x \frac{d^2y}{dx^2} + (\sin x)\frac{dy}{dx} + (1+x)y = \cot x$ is a linear D.E. in y

(2) $y'' = 3yy'$ & $\left(\frac{d^2y}{dx}\right)^2 + 3y = 0$ are non-linear.

- Differential eq^n (1) is said to be non homogeneous if the forcing term $R(x) \neq 0$.

- otherwise it is said to be homogeneous.

$$P_0(x)\frac{d^n y}{dx^n} + P_1(x)\frac{d^{n-1}y}{dx^{n-1}} + P_2(x)\frac{d^{n-2}y}{dx^{n-2}} + \dots + P_n(x)y = 0 \quad \text{---(2)}$$

is known as homogeneous linear differential eq^n.

* Operator Form of Differential eq^n:

If we represent the differential operator $\frac{d}{dx}$ by D then

$$\frac{d}{dx} = D, \quad \frac{d^2}{dx^2} = D^2, \quad \dots, \quad \frac{d^n}{dx^n} = D^n.$$

Scanned with CamScanner

∴ Eq^{n} ① reduces to

$$P_0(x) D^n y + P_1(x) D^{n-1} y + P_2(x) D^{n-2} y + \cdots + P_n(x) y = R(x)$$

OR

$$(P_0(x) D^n + P_1(x) D^{n-1} + P_2(x) D^{n-2} + \cdots + P_n) y = R(x)$$

OR

$$[f(D)] y = R(x) \quad \text{——} \quad (3)$$

where $f(D)$ represent polynomial in D.

- Similarly, the homogeneous eq^{n} ② may be written in the operator form as

$$[f(D)] y = 0 \quad \text{——} \quad (4)$$

Scanned with CamScanner

* Second order Homogeneous Equations with constant coefficients:-

An ordinary differential eq^{n} of the form

$$a_0\frac{d^n y}{dx^n} + a_1\frac{d^{n-1}y}{dx^{n-1}} + a_2\frac{d^{n-2}y}{dx^{n-2}} + \cdots + a_n y = 0 \quad \text{---(1)}$$

where $a_0, a_1, a_2, \cdots, a_n$ are constants, is known as a homogeneous linear differential equation of order n with constant coefficients, where $a_0 \neq 0$.

This eq^{n} is known as linear since the degree of the dependent variable y and all its differential co-efficients is one.

Eq^{n} (1) can also written as

$$(a_0D^n + a_1D^{n-1} + a_2D^{n-2} + \cdots + a_n)y = 0 \quad \text{---(2)}$$

$\therefore f(D)y = 0$, where $f(D) = a_0D^n + a_1D^{n-1} + \cdots + a_n$

Here $D = \frac{d}{dx}$ is known as the differential operator.

second order homogeneous eq^{ns} with constant co-efficients.

$$a_0\frac{d^2y}{dx^2} + a_1\frac{dy}{dx} + a_2y = 0, \quad \text{where } a_0 \neq 0.$$

General Solution of a Homogeneous Linear Differential Equation:-

Assume that $y = e^{mx}$ be the sol^{n} of eq^{n} (2), where m is a constant (real or complex.)

$\therefore Dy = me^{mx}$, $D^2y = m^2e^{mx}$, $\cdots$, $D^ny = m^ne^{mx}$.

Substituting these values in eq^{n} (2),

Scanned with CamScanner

$(a_0 e^{mx} \cdot m^n + a_1 m^{n-1} e^{mx} + a_2 m^{n-2} e^{mx} + \dots + a_n e^{mx}) = 0$

$\therefore e^{mx}(a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n) = 0$

Since, $e^{mx} \neq 0$

$\therefore a_0 m^n + a_1 m^{n-1} + a_2 m^{n-2} + \dots + a_n = 0$ ——— (3)

Eq^n (3) is called Auxiliary or characteristic-eq^u of eq^n (1).

$\therefore f(m) = 0$ is called A.E.

- If $m_1, m_2, m_3, \dots, m_n$ are n roots of the auxiliary eq^n, the following cases arises.

Case-I:- Real and Distinct Roots:-

If roots $m_1, m_2, m_3, \dots, m_n$ are real and distinct then the sol^n of eq^n (1) is given by

$y = c_1 e^{m_1 x} + c_2 e^{m_2 x} + c_3 e^{m_3 x} + \dots + c_n e^{m_n x}$

e.g. Solve $y'' + 3y' + 2y = 0$; $y' = \frac{dy}{dx}$.

- Rewritting the D.E. in operator form,

$(D^2 + 3D + 2)y = 0$

Therefore A.E. is $m^2 + 3m + 2 = 0$

$\therefore (m+1)(m+2) = 0$

$\therefore m = -1, -2.$

$m_1 = -1, \; m_2 = -2$

which are real & distinct.

$\therefore$ The General solution is

$y = c_1 e^{-x} + c_2 e^{-2x}$, where c_1 & c_2 are arbitrary constants.

Scanned with CamScanner

Case-I When all the n-roots of the A.E. are real and equal.

Let, $m_1, m_2, \ldots, m_n$ be the n real root of A.E. and $m_1 = m_2 = \ldots = m_n = m$ (say) then the G.S. is

$y = (C_1 + C_2 x + C_3 x^2 + \ldots + C_n x^{n-1}) e^{mx}$, where $C_1, C_2, \ldots, C_n$ are arbitrary constants.

e.g. Solve $\frac{d^3y}{dx^3} - 3\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - y = 0$

$\Rightarrow (D^3 - 3D^2 + 3D - 1) y = 0$

$\therefore (D-1)^3 y = 0$ A.E. is $m^3 - 3m^2 + 3m - 1 = 0$

$\therefore m = 1, 1, 1.$ $\rightarrow$ Real & repeated roots.

$\therefore$ The G.S. is $y = (C_1 + C_2 x + C_3 x^2) e^x$.

Corollary :- When all the n-roots of the A.E. are real but some of them are repeated.

Let, $m_1, m_2, \ldots, m_n$ be the n real roots of the A.E. and $m_1 = m_2 = \ldots = m_k$ & $m_{k+1} \neq m_{k+2} \neq \ldots \neq m_n$ then the G.S. is

$y = (C_1 + C_2 x + C_3 x^2 + \ldots + C_k x^{k-1}) e^{mx} + C_{k+1} e^{m_{k+1} x} + C_{k+2} e^{m_{k+2} x} + \ldots + C_n e^{m_n x}$.

e.g. Solve $(D^4 - D^3 - 6D^2 + 4D + 8) y = 0$

A.E. is $m^4 - m^3 - 6m^2 + 4m + 8 = 0$ $(m+1)$ is one of the factor. Can be solved by that method too.

$\therefore m^4 + m^3 - 2m^3 - 2m^2 - 4m^2 + 4m + 8m + 8 = 0$

$\therefore m^3(m+1) - 2m^2(m+1) - 4m(m+1) + 8(m+1) = 0$

$\therefore (m+1)(m^3 - 2m^2 - 4m + 8) = 0$

$\therefore (m+1)[m^2(m-2) - 4(m-2)] = 0$

$\therefore (m+1)(m-2)(m^2 - 4) = 0$

$\therefore (m+1)(m-2)(m-2)(m+2) = 0$

Scanned with CamScanner

$\therefore (m+1)(m-2)^2(m+2) = 0$

$\therefore m = -1, -2, 2$ (Repeated 2 times).

$\therefore$ The G.S. is $y = C_1 e^{-x} + C_2 e^{-2x} + (C_3 + C_4 x) e^{2x}$

* Case - 3:- When all the n-roots of the A.E. are non-repeated complex conjugate pairs.

Let $\alpha_1 \pm i\beta_1, \alpha_2 \pm i\beta_2, \ldots, \alpha_n \pm i\beta_n$ be the n-roots of A.E. which are non-repeated complex pairs, then the G.S. is

$y = e^{\alpha_1 x}(C_1 \cos\beta_1 x + C_2 \sin\beta_1 x) + e^{\alpha_2 x}(C_3 \cos\beta_2 x + C_4 \sin\beta_2 x)$

$+ \ldots + e^{\alpha_n x}(C_{n-1}\cos\beta_n x + C_n \sin\beta_n x)$.

e.g. Solve $(D^4 + 4D^3 + 9D^2 + 10D + 6)y = 0$.

$\rightarrow$

A.E. is $m^4 + 4m^3 + 9m^2 + 10m + 6 = 0$

$\therefore$ ~~$m^2(m^2 + 2m +$~~

$\therefore m^4 + 2m^3 + 2m^3 + 3m^2 + 4m^2 + 2m^2 + 6m + 4m + 6 = 0$

$\therefore m^4 + 2m^3 + 3m^2 + 2m^3 + 4m^2 + 6m + 2m^2 + 4m + 6 = 0$

$\therefore m^2(m^2 + 2m + 3) + 2m(m^2 + 2m + 3) + 2(m^2 + 2m + 3) = 0$

$\therefore (m^2 + 2m + 3)(m^2 + 2m + 2) = 0$

$\therefore m^2 + 2m + 3 = 0$ or $m^2 + 2m + 2 = 0$

$m = \frac{-2 \pm \sqrt{4 - 4 \cdot 1 \cdot 3}}{2}$ or $m = \frac{-2 \pm \sqrt{4 - 4 \cdot 1 \cdot 2}}{2}$

$m = \frac{-2 \pm \sqrt{-8}}{2}$ or $m = \frac{-2 \pm \sqrt{-4}}{2}$

$\therefore m = \frac{-2 \pm i2\sqrt{2}}{2}$ or $m = \frac{-2 \pm 2i}{2}$

Scanned with CamScanner

$\therefore m = -1 \pm \sqrt{2}i$ or $m = -1 \pm i$.

The roots are of the form $\alpha_1 \pm i\beta_1, \alpha_2 \pm i\beta_2$.

The G.S. is

$y = e^{-x}(C_1 \cos\sqrt{2}x + C_2 \sin\sqrt{2}x) + e^{-x}(C_3 \cos x + C_4 \sin x)$

* Case-4 :- When all the n-roots of the A.E. are repeated complex conjugate pairs.

Let, $\alpha \pm i\beta$ be the n-roots of the A.E. which are repeated complex conjugate pairs. then the G.S. is,

$y = e^{\alpha x}[(C_1 + C_2 x + \ldots + C_k x^{k-1})\cos\beta x + (C_{k+1} + C_{k+2} x + \ldots + C_n x^{n-k})\sin\beta x]$.

e.g. Solve $(D^2+2D+2)^2 y = 0$

$\Rightarrow$

A.E. is $(m^2+2m+2)^2 = 0$

By previous example, $m = -1 \pm i$ (Repeated 2 times)

$\therefore$ The G.S. is $y = e^{-x}[(C_1 + C_2 x)\cos x + (C_3 + C_4 x)\sin x]$.

* Note :-

If two pairs of complex roots m_1, m_2 and m_3, m_4 are equal, say $m_1 = m_2 = \alpha + i\beta$, $m_3 = m_4 = \alpha - i\beta$ and remaining $(n-4)$ roots $m_5, m_6, \ldots, m_n$ are real & distinct then sol^n is

$y = e^{\alpha x}[(C_1 + C_2 x)\cos\beta x + (C_3 + C_4 x)\sin\beta x] + C_5 e^{m_5 x} + C_6 e^{m_6 x} + \ldots + C_n e^{m_n x}$.

Scanned with CamScanner

*Note:-

In the C.S. of a homogeneous eqⁿ, the number of arbitrary constants is always equal to the order of that homogeneous eqⁿ.

e.g. of Note (1):-

Solve $y''' - y'' + 100y' - 100y = 0$, $y(0) = 4$, $y'(0) = 11$, $y''(0) = -299$.

→

In operator form, the given D.E. is

$m^3 - m^2 + 100m - 100 = 0$

$\therefore m^2(m-1) + 100(m-1) = 0$

$\therefore (m-1)(m^2+100) = 0$

$\therefore m = 1, \quad m = \pm 10i$

↓ Real ↓ Non repeated complex pair of roots

- The C.S. is

$y = C_1 e^x + e^{0x}(C_2 \cos 10x + C_3 \sin 10x)$ —(1)

- When $y(0) = 4$

$\therefore 4 = C_1 + C_2$ —(2)

- $y' = C_1 e^x - C_2 10 \sin 10x + 10 C_3 \cos 10x$

$y'' = C_1 e^x - 100 C_2 \cos 10x - 100 C_3 \sin 10x$

- $y'(0) = 11$.

$\therefore 11 = C_1 + 100 C_3$ —(3)

$y''(0) = -299$

$\therefore -299 = C_1 - 100 C_2$ —(4)

Scanned with CamScanner

Multiply (4) by (-1),

$\therefore 299 = -C_1 + 100C_3$

$4 = C_1 + C_2$

$303 = 101C_2 + C_2$

Multiply eq^n (1) by (-1)

$\therefore 299 = -C_1 + 100C_3$

From (2), $-4 = -C_1 - C_2$

$-299 = C_1 - 100C_2$

$-303 = -101C_2.$

$\therefore \boxed{C_2 = 3}$

$\boxed{C_1 = 1}$ From (2),

From (3), $\boxed{C_3 = 1}$

∴ Particular sol^n of the given D.E. is

$y = e^x + 3\cos 10x + \sin 10x.$

* Examples :-

① Solve $2D^2y + Dy - 6y = 0$

→

The eq^n can be written as

$(2D^2 + D - 6)\, y = 0$

The Auxiliary eq^n is $2m^2 + m - 6 = 0$

$2m^2 + 4m - 3m - 6 = 0$

$\therefore 2m(m+2) - 3(m+2) = 0$

$\therefore (m+2)(2m-3) = 0$

$\therefore m = -2, \frac{3}{2}$ (Case-I)

Real & distinct.

The G.S. is $y = C_1e^{-2x} + C_2e^{3/2x}$

Scanned with CamScanner

(2) Solve $\frac{d^2x}{dt^2} + 6\frac{dx}{dt} + 9x = 0$

→

D.E. - $(D^2+6D+9)x = 0$

A.E. is $m^2+6m+9=0$

$\therefore (m+3)^2 = 0$

$\therefore m = -3, -3$ (Case - II) Real & Repeated

Hence, the G.S. is $y = e^{-3x}(C_1 + C_2 t)$.

(3) Solve $y'' + 4y' + 4y = 0$, $y(0) = 1$, $y'(0) = 1$.

→

A.E. is $(m^2+4m+4) = 0$

$\therefore (m+2)^2 = 0$

$m = -2, -2$. Real & repeated

- The G.S. is

$$y = e^{-2x}(C_1 + C_2 x) \quad \text{—— (1)}$$

Differentiating eq^n (1) by

$$y' = -2e^{-2x}(C_1 + C_2 x) + C_2 e^{-2x} \quad \text{—— (2)}$$

$y(0) = 1$, From (1),

$$1 = C_1 \quad \text{—— (3)}$$

$y'(0) = 1$, From (2),

$$1 = -2C_1 + C_2$$

$\therefore C_2 = 3$

$\therefore$ The particular sol^n is $y = e^{-2x}(1+3x)$

Scanned with CamScanner

* Superposition principle or Linearity principle or fundamental Theorem for Linear homogeneous Differential Eq^R:

If $y_1(x), y_2(x), \ldots, y_n(x)$ are n solutions of homogeneous linear differential eq^n (2) then the linear combination $c_1y_1(x) + c_2y_2(x) + \cdots + c_ny_n(x)$ is also a solution of eq^R (2), where $c_1, c_2, \ldots, c_n$ are arbitrary constants.

* Note:-
This thm. does not hold for nonhomogeneous linear D.E.

should start method of reduction of order

* Initial and boundary value problem:-

- When the solution of a D.E. is obtained subject to the conditions on the unknown functions and its derivatives specified at one value of the independent variable, then the problem is called I.V.P.

Scanned with CamScanner

- A differential eqⁿ together with an initial condition is called an initial value problem. with x as the independent variable.
- It is of the form

$$y' = f(x, y), \quad y(x_0) = y_0.$$

e.g. Find the curve through the point (1,1) in the xy-plane having at each of its points the slope $-y/x$.

⇒

Here, $y' = -\frac{y}{x}$.

$$\therefore \frac{dy}{dx} = \frac{-y}{x}$$

$$\therefore \frac{1}{y}dy = \frac{-1}{x}dx$$

$$\therefore \log y = -\log x + \log c$$

$$\therefore \log x + \log y = \log c$$

$$\therefore xy = c$$

Given $x = 1, \ y = 1$

$$\therefore c = 1$$

$\therefore y = \frac{1}{x}$ is the particular solⁿ.

e.g. I.V.P $y'' - y = 0$; $y' = \frac{dy}{dx}$, $y(0) = 4$, $y'(0) = -2$.

- Let, $y_1 = e^x$ & $y_2 = e^{-x}$ are two solⁿ of the given D.E.

$$\therefore y = c_1 e^x + c_2 e^{-x}$$

Take from the initial conditions

$$y(0) = c_1 + c_2 = 4$$

$$y' = c_1 e^x - c_2 e^{-x}, \quad y'(0) = c_1 - c_2 = -2$$

$$\therefore c_1 = 1, \ c_2 = 3.$$

Scanned with CamScanner

(4) Solve the initial value problem $y''-4y'+4y=0$, $y(0)=3$, $y'(0)=1$.

→

The auxiliary eq^n is $m^2-4m+4=0$

$\therefore (m-2)^2=0$

$\therefore m=2,2$ (real & repeated)

$\therefore$ The G.S. is $y=e^{2x}(C_1+C_2x)$ —— (1)

Diff eq^n (1) w.r.to x, $y'=e^{2x}C_2+2e^{2x}(C_1+C_2x)$ —— (2)

Now, $y(0)=3$

$\therefore$ From (1), $3=C_1$ —— (3)

Given that $y'(0)=1$

$\therefore$ From (2), $1=C_2+2C_1$

From (3), $C_2=-5$

Hence, The particular sol^n is $y=e^{2x}(3-5x)$.

(5) Solve the I.V.P. $y''-9y=0$, $y(0)=2$, $y'(0)=-1$.

→

The A.E. is $m^2-9=0$

$(m+3)(m-3)=0$

$\therefore m=-3,3$ (Real & distinct.)

Hence, the G.S. is $y=C_1e^{-3x}+C_2e^{3x}$. —— (1)

Diff. eq^n (1) w.r.to x, $y'=-3C_1e^{-3x}+3C_2e^{3x}$. —— (2)

Given that $y(0)=2$.

$\therefore$ From (1), $2=C_1+C_2$ —— (3)

Scanned with CamScanner

Also, given that $y'(0) = -1$.

$\therefore$ From (2), $-1 = -3C_1 + 3C_2$ ——— (4)

$[3 \times (3)] + (4)$,

$$3C_1 + 3C_2 = 6$$
$$-3C_1 + 3C_2 = -1$$
$$6C_2 = 5$$
$$C_2 = \frac{5}{6}$$

From eq^n (3), $C_1 + \frac{5}{6} = 2$

$$\therefore C_1 = 2 - \frac{5}{6} = \frac{7}{6}.$$

$\therefore$ The particular sol^n is $y = \frac{7}{6} e^{-3x} + \frac{5}{6} e^{3x}$.

(6) Solve $(D^3 - 5D^2 + 8D - 4)y = 0$

→

The A.E. is $m^3 - 5m^2 + 8m - 4 = 0$

$m^3 - m^2 - 4m^2 + 4m + 4m - 4 = 0$

$\therefore m^2(m-1) - 4m(m-1) + 4(m-1) = 0$

$\therefore (m-1)(m^2 - 4m + 4) = 0$

$\therefore (m-1)(m-2)^2 = 0$

$\therefore m = 1, 2, 2$

$\therefore$ The G.S. is $y = C_1 e^x + e^{2x}(C_2 + C_3 x)$.

(7) Solve $(D^3 + 1)y = 0$

→

The A.E. is $m^3 + 1 = 0$

$\therefore (m+1)(m^2 - m + 1) = 0$

$\therefore m + 1 = 0, \quad m^2 - m + 1 = 0$

$\therefore m = -1, \quad m = \frac{+1 \pm \sqrt{1+4}\, i}{2}$

Scanned with CamScanner

$\therefore m = -1, \quad m = \frac{+1 \pm \sqrt{3}i}{2} = \frac{1}{2} \pm \frac{\sqrt{3}}{2}i$

$\therefore y = C_1 e^{-x} + e^{\frac{1}{2}x}\left(C_2 \cos\frac{\sqrt{3}}{2}x + C_3 \sin\frac{\sqrt{3}}{2}x\right)$

(8) Solve $(D^4 - 2D^3 + D^2)y = 0$

→

A.E is $(m^4 - 2m^3 + m^2) = 0$

$\therefore m^2(m^2 - 2m + 1) = 0$

$\therefore m^2(m-1)^2 = 0$

$\therefore m = 0, 0, 1, 1$

$\therefore y = e^{0x}(C_1 + C_2 x) + e^{x}(C_3 + C_4 x)$

(9) Solve $(D^4 - 1)y = 0$

→

A.E is $m^4 - 1 = 0$

$(m^2-1)(m^2+1) = 0$

$(m+1)(m-1)(m^2+1) = 0$

$\therefore m = \pm 1, \quad m^2 = -1 = \pm i^2$

$\therefore y = C_1 e^{x} + C_2 e^{-x} + e^{0x}(C_3 \cos x + C_4 \sin x)$

(10) Solve $(D^4 + 4)y = 0$

A.E. is $m^4 + 4 = 0$

$\therefore m^4 + 4 + 4m^2 - 4m^2 = 0$

$\therefore (m^2+2)^2 - (2m)^2 = 0$

$\therefore (m^2 + 2 + 2m)(m^2 + 2 - 2m) = 0$

$\therefore (m^2 + 2m + 2)(m^2 - 2m + 2) = 0$

$\therefore m = \frac{-2 \pm \sqrt{4}i}{2}, \quad m = \frac{2 \pm \sqrt{4}i}{2}$

$\therefore m = -1 \pm i, \quad m = 1 \pm i$

Scanned with CamScanner

$\therefore y = e^{-x}(C_1 \cos x + C_2 \sin x) + e^{x}(C_3 \cos x + C_4 \sin x)$.

(11) Solve $(D^4 + 8D^2 + 16)y = 0$

→

A.E. is $(m^4 + 8m^2 + 16) = 0$

$\therefore (m^2 + 4)^2 = 0$

$\therefore m^2 + 4 = 0, \quad m^2 + 4 = 0$

$\therefore m^2 = -4, \quad m^2 = -4$

$\therefore m = \pm 2i, \quad m = \pm 2i$

$\therefore y = e^{0x}[(C_1 + C_2 x)\cos 2x + (C_3 + C_4 x)\sin 2x)$.

(12) $(D^4 + 4D^2)y = 0$

→

A.E is $m^4 + 4m^2 = 0$

$m^2(m^2 + 4) = 0$

$m^2 = 0, \quad m^2 + 4 = 0$

$m = 0, 0, \quad m^2 = -4 = \pm 2i$

$\therefore y = e^{0x}(C_1 + C_2 x) + e^{0x}(C_3 \cos 2x + C_4 \sin 2x)$

$= C_1 + C_2 x + C_3 \cos 2x + C_4 \sin 2x$.

Scanned with CamScanner

* Non-Homogeneous Linear Differential Equations of Higher order with constant co-efficients

Any ordinary differential eq^{n} of the form

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \cdots + a_n y = Q(x) \quad \text{---(1)}$$

where $a_0, a_1, a_2, \ldots, a_n$ are constants and Q is a function of x, is known as a non-homo L.D.E. with constant co-efficients.

$$(a_0 D^n + a_1 D^{n-1} + a_2 D^{n-2} + \cdots + a_n) y = Q(x) \quad \text{---(2)}$$

$$f(D)\, y = Q(x).$$

where, $f(D) = a_0 D^n + a_1 D^{n-1} + \cdots + a_n$

General solution of a Non homogeneous Linear Differential Equation:-

A general solution of eq^{n} (1) is obtained in two parts as

G.S = Complementary Function + Particular Integral

$$y = C.F. + P.I$$

$$y = y_c(x) + y_p(x) \quad \text{or} \quad y = y_h(x) + y_p(x)$$

Scanned with CamScanner

- The complementary function (CF) is the general solution of the homogeneous equation obtained by putting $Q(x) = 0$ in eq^{n} (1).

- The particular integral (PI) is any particular solution of the non homogeneous eq^{n} (1) and contains no arbitrary constants.

* Inverse operator and Particular Integral

- $f(D)$ is known as the differential operator and $\frac{1}{f(D)}$ is known as the inverse differential operator.

$$f(D)\left[\frac{1}{f(D)} Q(x)\right] = Q(x).$$

- This shows that $\frac{1}{f(D)} Q(x)$ satisfies the equation $f(D)y = Q(x)$ and since $\frac{1}{f(D)} Q(x)$ does not conta any arbitrary constants, it gives the P.I. of the eq^{n} $f(D)y = Q(x)$.

- Hence, $P.I. = \frac{1}{f(D)} Q(x)$.

i) If $f(D) = D$ then

$$P.I. = \frac{1}{D} Q(x) = \int Q(x)\, dx.$$

ii) If $f(D) = D - a$ then the eq^{n} $f(D)y = Q(x)$ becomes

$$(D-a)y = Q(x)$$

$$\frac{dy}{dx} - ay = Q(x)$$

which is first order L.D.E.

Scanned with CamScanner

$I.F. = e^{\int -a\,dx} = e^{-ax}$

∴ The sol^n is $y e^{-ax} = \int Q(x) e^{-ax} dx + C$.

$\therefore y = e^{ax} \int Q(x) e^{-ax} dx + C e^{ax}$.

Here, Ce^{ax} is the complementary function since it contains the arbitrary constant c and $e^{ax} \int Q(x) e^{-ax} dx$ is the particular integral.

Hence, $P.I. = \frac{1}{D-a} Q(x) = e^{ax} \int Q(x) e^{-ax} dx$. —(1)

* ~~Direct~~ (short-cut) ~~method of obtaining particular integrals.~~

~~Method 1~~ :-

When $f(D) = (D-a_1)(D-a_2)\ldots(D-a_n)$. then we have two methods to find P.I.

Method 1 :-

$$\frac{1}{(D-a_1)(D-a_2)\cdots(D-a_n)} Q(x)$$

$$= \frac{1}{(D-a_1)(D-a_2)\cdots(D-a_{n-1})} \left[\frac{1}{D-a_n} Q(x)\right]$$

$$= \frac{1}{(D-a_1)(D-a_2)\cdots(D-a_{n-1})} e^{a_n x} \int Q(x) e^{-a_n x} dx. \quad \text{(From (1))}$$

$$= \frac{1}{(D-a_1)(D-a_2)\cdots(D-a_{n-1})} (\text{Function of } x)$$

$$= \frac{1}{(D-a_1)(D-a_2)\cdots(D-a_{n-2})} \left[\frac{1}{(D-a_{n-1})} (\text{Function of } x)\right].$$

Scanned with CamScanner

- It will continue till we operator the last operator $\frac{1}{D-a_1}$ on the last obtained function of x.

Method 2:- Partial Fractions Decomposition

$$\frac{1}{(D-a_1)(D-a_2)\cdots(D-a_n)} Q(x) = \left[\frac{A_1}{D-a_1} + \frac{A_2}{D-a_2} + \cdots + \frac{A_n}{D-a_n}\right] Q(x)$$

Then apply eq^n (1) to each of this term.

- The above methods to obtained P.I. are known as general methods.

* Examples:-

(1) Find the G.S. of $(D-3)y = xe^{2x}$.

→

C.F. is the general sol^n of $(D-3)y=0$

A.E. is $m-3=0$

$\therefore m = 3$

$\therefore C.F. = Ce^{3x}$

P.I. or y_p or $y_h = \frac{1}{f(D)} Q(x)$

$$= \frac{1}{D-3} xe^{2x}$$

$$= e^{3x} \int xe^{2x} e^{-3x}\, dx$$

$$= e^{3x} \int xe^{-x}\, dx \qquad u = x, \quad v = e^{-x}$$

$$= e^{3x} \left[-xe^{-x} - \int (-e^{-x})\, dx\right]$$

$$= e^{3x} \left[-xe^{-x} + e^{-x}\right] dx$$

$$= e^{3x} \left[-e^{-x}(x+1)\right] = -e^{2x}(x+1).$$

Scanned with CamScanner

The G.S. of given D.E. is $y = C.F. + P.I.$

$= Ce^{3x} - (x+1)e^{2x}$.

(2) Find the P.I. of $(D^2 - 3D + 2)y = xe^x$

→

$$y_p = \frac{1}{f(D)} \phi(x)$$

$$= \frac{1}{(D^2 - 3D + 2)} xe^x$$

$$= \frac{1}{(D-1)(D-2)} xe^x$$

Method 1:-

$$y_p = \frac{1}{(D-1)(D-2)} xe^x$$

$$= \frac{1}{D-1}\left[\frac{1}{D-2} xe^x\right]$$

$$= \frac{1}{D-1}\left[e^{2x}\int xe^x e^{-2x}\,dx\right]$$

$$= \frac{1}{D-1}\left[e^{2x}\int xe^{-x}\,dx\right]$$

$$= \frac{1}{D-1}\left[e^{2x}\{-e^{-x}(x+1)\}\right]$$

$$= \frac{1}{D-1}\left[-(1+x)e^x\right]$$

$$= -e^x \int (1+x)e^x e^{-x}\,dx$$

$$= -e^x \int (1+x)\,dx$$

$$= -e^x\left[x + \frac{x^2}{2}\right] = \frac{-1}{2}(x^2+2x)e^x \quad \text{— (A)}$$

Scanned with CamScanner

Method 2:-

$$y_p = \frac{1}{(D-1)(D-2)} xe^x$$

$$= \left[\frac{1}{D-2}\right] \text{ By partial fraction method,}$$

$$\frac{1}{(D-1)(D-2)} = \frac{A}{D-1} + \frac{B}{D-2} \quad \text{---(1)}$$

$$\therefore 1 = A(D-2) + B(D-1)$$

Take $D=2$, $1 = B$

Take $D=1$, $1 = -A$

$\therefore A = -1$

In eq^n (1),

$$\frac{1}{(D-1)(D-2)} = \frac{-1}{D-1} + \frac{1}{D-2}$$

$$\therefore y_p = \left[\frac{-1}{D-1} + \frac{1}{D-2}\right] xe^x$$

$$= \left[\frac{1}{D-2} - \frac{1}{D-1}\right] xe^x$$

$$= \frac{1}{D-2} xe^x - \frac{1}{D-1} xe^x$$

$$= e^{2x} \int xe^x e^{-2x} dx - e^x \int xe^x e^{-x} dx$$

$$= e^{2x} \int xe^{-x} dx - e^x \int x \, dx$$

$$= e^{2x} \left[-e^{-x}(x+1)\right] - e^x \frac{x^2}{2}$$

$$= -e^x(x+1) - \frac{x^2}{2} e^x$$

$$= -e^x x - e^x - e^x \frac{x^2}{2} = \frac{-2e^x x - 2e^x - e^x x^2}{2}$$

Scanned with CamScanner

$\therefore y_p = \frac{-1}{2} e^x (2x + 2 + x^2)$ ——— (B)

- Eq^n (A) & (B) should be same but they are not. Eq^n (B) has the additional term $-e^x$.

- This term gets absorbed in the term $C_1 e^x$ of the C.F. $y_c = C_1 e^x + C_2 e^{2x}$ of the D.E. due to arbitrariness of C_1.

(3) Solve $(D^2 + 3D + 2)y = e^{e^x}$.

→

A.E. is $m^2 + 3m + 2 = 0$

$(m+2)(m+1) = 0$

$m = -1, -2$.

$\therefore$ C.F. $= C_1 e^{-x} + C_2 e^{-2x}$

P.I. $= \frac{1}{D^2 + 3D + 3} e^{e^x}$

$= \frac{1}{(D+2)(D+1)} e^{e^x}$

$= \frac{1}{D+2} \left[\frac{1}{D+1} e^{e^x} \right]$

$= \frac{1}{D+2} \left[e^{-x} \int e^{e^x} e^x \, dx \right]$

Putting $e^x = t$

$e^x dx = dt$

$= \frac{1}{D+2} \left[e^{-x} \int e^t \, dt \right]$

$= \frac{1}{D+2} e^{-x} e^t$

Scanned with CamScanner

$$P.I. = \frac{1}{D+2} e^{-2x} e^{e^x}$$

$$= e^{-2x} \int e^{-x} e^{e^x} e^{2x} dx$$

$$= e^{-2x} \int e^{x} e^{e^x} dx$$

$$= e^{-2x} e^{e^x}$$

The G.S. is $y = c_1 e^{-x} + c_2 e^{-2x} + e^{-2x} e^{e^x}$.

* Shorter methods for finding P.I. for special Form of Q(x) :-

- This method depends on the nature of eq^{n} (1)
- P.I. by this method can be obtained when Q(x) has the following forms.

1) $Q(x) = e^{ax+b}$
2) $Q(x) = \sin(ax+b)$ or $\cos(ax+b)$
3) $Q(x) = x^m$ or polynomial in x.
4) $Q(x) = e^{ax} V(x)$
5) $Q(x) = x \cdot V(x)$

Case-I :- $Q(x) = e^{ax+b}$ or e^{ax}

$$f(D)y = e^{ax+b}$$

Now, $D(e^{ax+b}) = ae^{ax+b}$, $D^2(e^{ax+b}) = a^2 e^{ax+b}$, ...,
$D^n(e^{ax+b}) = a^n e^{ax+b}$.

Consider $f(D)(e^{ax+b}) = (a_0 D^n + a_1 D^{n-1} + \dots + a_n) e^{ax+b}$

$$= (a_0 a^n e^{ax+b} + a_1 a^{n-1} e^{ax+b} + \dots + a_n e^{ax+b})$$

$$= (a_0 a^n + a_1 a^{n-1} + \dots + a_n) e^{ax+b}$$

$$= f(a) e^{ax+b}$$

Scanned with CamScanner

operating both the sides with $\frac{1}{f(D)}$.

$$\frac{1}{f(D)}\left[f(D)\, e^{ax+b}\right] = \frac{1}{f(D)}\left[f(a)\, e^{ax+b}\right]$$

$$\therefore e^{ax+b} = f(a)\,\frac{1}{f(D)}\, e^{ax+b}$$

$$\therefore \frac{1}{f(a)}\, e^{ax+b} = \frac{1}{f(D)}\, e^{ax+b}, \quad f(a) \neq 0.$$

$$\therefore \frac{f}{f(D)}\, e^{ax+b} = \frac{1}{f(a)}\, e^{ax+b}, \quad f(a) \neq 0.$$

Hence, $P.I = \frac{1}{f(a)}\, e^{ax+b}$ if $f(a) \neq 0$

* Note!-

If $f(a) = 0$ then $(D-a)$ is a factor of $f(D)$ and hence the above result fails.

Let, $f(D) = (D-a)\,\phi(D)$, where $\phi(a) \neq 0$.

$$P.I. = \frac{1}{f(D)}\, e^{ax+b}$$

$$= \frac{1}{(D-a)\,\phi(D)}\, e^{ax+b}$$

$$= \frac{1}{\phi(a)}\,\frac{1}{D-a}\, e^{ax+b}$$

$$= \frac{1}{\phi(a)}\, e^{ax} \int e^{ax+b}\, e^{-ax}\, dx$$

$$= \frac{1}{\phi(a)}\, e^{ax} \int e^{b}\, dx$$

Scanned with CamScanner

P.I $= \frac{1}{\phi(D)} e^{ax} \cdot x e^{b}$

$= e^{ax+b} \cdot \frac{1}{\phi(a)}$ ——— (1)

Since, $f(D) = (D-a)\phi(D)$

$f'(D) = (D-a)\phi'(D) + \phi(D)$

$f'(a) = 0 + \phi(a)$

$\therefore f'(a) = \phi(a)$.

Substituting in eq^{n} (1).

$\frac{1}{f(D)} e^{ax+b} = x \cdot \frac{1}{f'(a)} e^{ax+b}$, where $f'(a) \neq 0$.

- If $f'(a) = 0$ then repeating the above process,

$\frac{1}{f(D)} e^{ax+b} = x \cdot \left[x \cdot \frac{1}{f''(a)} e^{ax+b} \right]$

$= x^2 \cdot \frac{1}{f''(a)} e^{ax+b}$, where $f''(a) \neq 0$

- In general, if $(D-a)^r$ is a factor of $f(D)$, then

$\frac{1}{f(D)} e^{ax} = x^r \frac{1}{f^{r}(a)} e^{ax+b}$.

Hence, P.I $= x^r \frac{1}{f^{(r)}(a)} e^{ax+b}$.

*** Examples:-**

(1) Solve $(4D^2 - 4D + 1)y = 4$.

Scanned with CamScanner

→

The A.E. is $4m^2 - 4m + 1 = 0$, $4m^2 - 2m - 2m + 1 = 0$

$(2m-1)^2 = 0$ $\quad 2m(2m-1) - 1(2m-1) = 0$

$m = \frac{1}{2}, \frac{1}{2}$

$\therefore C.F. = e^{\frac{1}{2}x}(c_1 + c_2 x)$

$P.I. = \frac{1}{4D^2 - 4D + 1} 4e^{0x}$

$= 4 \frac{1}{4(0)^2 - 4(0) + 1} e^{0x}$

$= 4$

Hence, the G.S. is $y = (c_1 + c_2 x)e^{\frac{1}{2}x} + 4$.

3) Solve $(D^2 + 5D + 6)y = e^x$.

→

A.E. is $m^2 + 5m + 6 = 0$

$(m+2)(m+3) = 0$

$m = -2, -3$ (real & distinct).

$C.F. = c_1 e^{-2x} + c_2 e^{-3x}$

$P.I. = \frac{1}{D^2 + 5D + 6} e^x$

$= \frac{1}{1 + 5 + 6} e^x$

$= \frac{1}{12} e^x$.

Hence, the G.S. is $y = c_1 e^{-2x} + c_2 e^{-3x} + \frac{1}{12} e^x$.

Scanned with CamScanner

(3) Solve $(4D^2 - 4D + 1)y = e^{x/2}$.

→

The A.E. is $4m^2 - 4m + 1 = 0$

$(2m-1)^2 = 0$

$m = \frac{1}{2}, \frac{1}{2}$.

$\therefore$ C.F. $= e^{x/2}(c_1 + c_2 x)$

$P.I. = \frac{1}{4D^2 - 4D + 1} e^{x/2}$

$= x \cdot \frac{1}{8D - 4} e^{x/2}$

$= x^2 \cdot \frac{1}{8} e^{x/2}$

$\therefore$ The G.S. is $y = (c_1 + c_2 x) e^{x/2} + \frac{x^2}{8} e^{x/2}$.

(4) Solve $(D^2 - 4)y = e^{2x} + e^{-4x}$.

→

The A.E. is $m^2 - 4 = 0$

$(m-2)(m+2) = 0$

$\therefore m = 2, -2$.

$\therefore$ C.F. $= c_1 e^{2x} + c_2 e^{-2x}$.

$P.I. = \frac{1}{D^2 - 4}(e^{2x} + e^{-4x})$

$= \frac{1}{D^2 - 4} e^{2x} + \frac{1}{D^2 - 4} e^{-4x}$

$= x \cdot \frac{1}{2D} e^{2x} + \frac{1}{(-4)^2 - 4} e^{-4x}$

$= x \cdot \frac{1}{4} e^{2x} + \frac{1}{12} e^{-4x}$

Scanned with CamScanner

The G.S. is $y = C_1 e^{2x} + C_2 e^{-2x} + \frac{x}{4} e^{2x} + \frac{1}{12} e^{-4x}$.

(5) Solve $(D^2 + 4D + 5)y = -2\cosh x$.

→

- The A.E. is $m^2 + 4m + 5 = 0$

$\therefore m = \frac{-4 \pm \sqrt{16 - 20}}{2} = \frac{-4 \pm 2i}{2}$

$\therefore m = -2 \pm i$

$\therefore C.F. = e^{-2x}(C_1 \cos x + C_2 \sin x)$.

$P.I. = \frac{1}{D^2 + 4D + 5}(-2\cosh x)$

$= \frac{(-2)}{D^2 + 4D + 5}\left(\frac{e^x + e^{-x}}{2}\right)$.

$= -\frac{1}{D^2 + 4D + 5} e^x - \frac{1}{D^2 + 4D + 5} e^{-x}$

$= -\frac{1}{1 + 4 + 5} e^x - \frac{1}{(-1)^2 + 4(-1) + 5} e^{-x}$

$= -\frac{1}{10} e^x - \frac{1}{2} e^{-x}$

- The G.S. is $y = e^{-2x}(C_1 \cos x + C_2 \sin x) - \frac{1}{10} e^x - \frac{1}{2} e^{-x}$.

(6) Solve $(D^2 + 6D + 9)y = 5^x - \log 2$.

→

A.E. is $m^2 + 6m + 9 = 0$

$\therefore (m+3)^2 = 0$

$\therefore m = -3, -3$

$\therefore C.F. = (C_1 + C_2 x)e^{-3x}$.

Scanned with CamScanner

$$P.I. = \frac{1}{D^2+6D+9}(5^x - \log 2)$$

$$= \frac{1}{(D+3)^2} e^{x \log 5} - \frac{1}{(D+3)^2}(\log 2)\cdot e^{0x}$$

$$= \frac{5^x}{(\log 5+3)^2} - \log 2 \frac{1}{(0+3)^2} e^{0.x}$$

$$= \frac{5^x}{(\log 5+3)^2} - \frac{\log 2}{9}$$

- The G.S. is $y = (C_1 + C_2 x)e^{-3x} + \frac{5^x}{(\log 5+3)^2} - \frac{\log 2}{9}$.

(7) Solve $y''' - 3y'' + 3y' - y = 4e^t$.

→

The A.E. is $m^3 - 3m^2 + 3m - 1 = 0$

$(m-1)^3 = 0$

$\therefore m = 1, 1, 1$

$\therefore C.F. = e^t(C_1 + C_2 t + C_3 t^2)$

$$P.I. = \frac{1}{(D^3 - 3D^2 + 3D - 1)} 4e^t.$$

$$= \frac{1}{(D-1)^3} 4e^t$$

$$= 4t \frac{1}{3(D-1)^2} e^t$$

$$= \frac{4}{3} t^2 \frac{1}{2(D-1)} e^t$$

$$= \frac{2}{3} t^3 \frac{1}{1} e^t = \frac{2}{3} t^3 e^t.$$

Scanned with CamScanner

The G.S. is $y = e^t(C_1 + C_2 t + C_3 t^2) + \frac{2}{3} t^3 e^t$.

(5) solve $(D^3 - 12D + 16)y = (e^x + e^{-2x})^2$

→

$(D^3 - 12D + 16)y = (e^x + e^{-2x})^2$

$= e^{2x} + 2e^{-x} + e^{-4x}$

A.E. is $m^3 - 12m + 16 = 0$

$m^3 + 4m^2 - 4m^2 - 16m + 4m + 16 = 0$

$m^2(m+4) - 4m(m+4) + 4(m+4) = 0$

$(m+4)(m^2 - 4m + 4) = 0$

$(m+4)(m-2)^2 = 0$

$\therefore m = -4, 2, 2$

C.F. $= C_1 e^{-4x} + e^{2x}(C_2 + C_3 x)$

P.I. $= \frac{1}{D^3 - 12D + 16}(e^{2x} + 2e^{-x} + e^{-4x})$

$= \frac{1}{D^3 - 12D + 16} e^{2x} + 2\frac{1}{D^3 - 12D + 16} e^{-x} + \frac{1}{D^3 - 12D + 16} e^{-4x}$

$= \frac{1}{(D+4)(D-2)^2} e^{2x} + \frac{2}{(D+4)(D-2)^2} e^{-x} + \frac{1}{(D+4)(D-2)^2} e^{-4x}$

$= \frac{1}{(D-2)^2}\left[\frac{1}{D+4} e^{2x}\right] + \frac{2}{3 \cdot 9} e^{-x} + \frac{1}{D+4}\left[\frac{1}{(D-2)^2} e^{-4x}\right]$

$= \frac{1}{(D-2)^2} \frac{1}{6} e^{2x} + \frac{2}{27} e^{-x} + \frac{1}{D+4} \frac{1}{36} e^{-4x}$

$= \frac{1}{6} \frac{1}{(D-2)^2} e^{2x} + \frac{2}{27} e^{-x} + \frac{1}{36} \frac{1}{D+4} e^{-4x}$

$= \frac{1}{6} \frac{x \cdot 1}{2(D-2)} e^{2x} + \frac{2}{27} e^{-x} + \frac{1}{36} \frac{x \cdot 1}{1} e^{-4x}$

$= \frac{1}{6} \frac{x^2}{2} \frac{1}{1} e^{2x} + \frac{2}{27} e^{-x} + \frac{1}{36} x e^{-4x}$

Scanned with CamScanner

$$= \frac{x^2}{12} e^{2x} + \frac{2}{27} e^{-x} + \frac{x}{36} e^{-4x}.$$

- The G.S. is $y = C_1 e^{-4x} + (C_2 + C_3 x) e^{2x} + \frac{x^2}{12} e^{2x} + \frac{2}{27} e^{-x} + \frac{x}{36} e^{-4x}.$

Scanned with CamScanner